KING MONGKUT'S UNIVERSITY OF TECHNOLOGY THONBURI

DEPARTMENT OF COMPUTER ENGINEERING

CPE393: INTRODUCTION TO DATA SCIENCE WITH PYTHON

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PREDICTING HEART DISEASE RISK WITH MACHINE LEARNING

Final Group Project Report — SAMAI TEAM

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PROJECT DELIVERABLES

GitHub Repository : https://github.com/moigabone/CPE_Project_Samai_Team

Table of contents

1. Project Title and Institutional Framework

1.1 Project Title and Team Identity

- Project Title: Predicting Heart Disease Risk with Machine Learning
- Project Domain: Automated Clinical Decision Support Systems (CDSS)
- Development Group Name: SAMAI TEAM

1.2 Student Academic Roster

The technical research, algorithmic programming, and documentation pipeline presented within this comprehensive project report were executed entirely by the following group members :

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1.3 Course Instructors and Evaluation Framework

This final group project constitutes 50% of the total evaluation for Part 2 (Data Science with Python) within the academic syllabus of CPE393: Introduction to Data Science with Python. The work is submitted for formal grading to the academic faculty panel of the Computer Engineering Department at KMUTT.

1.4 Project Repositories and Academic Deliverables

Submission Date: 4 July 2026

Official Code Repository: https://github.com/moigabone/CPE_Project_Samai_Team

2. Problem Definition and Goals

2.1 Medical Context and Socio-Economic Motivation

Cardiovascular diseases (CVDs) remain the undisputed leading cause of human mortality on a global scale. According to statistical profiles updated by the World Health Organization (WHO), an estimated 17.9 million human lives are lost each year to cardiovascular complications. This figure accounts for approximately 31% of all global deaths. The economic burden of CVDs is equally staggering, placing immense pressure on healthcare infrastructure through long hospital stays, expensive surgical treatments, and chronic patient management.

The primary clinical problem with heart conditions is that they often develop silently over long periods. Pathological anomalies like coronary artery narrowing, lipid plaque accumulation, and subtle myocardial degradation frequently show no visible physical symptoms until an acute medical emergency—such as a sudden myocardial infarction (heart attack) or ischemic stroke—occurs.

Traditionally, defensive medicine relies on manual scoring systems like the Framingham Risk Score. While useful, these standard risk charts struggle to analyze the highly non-linear, multi-dimensional interactions that occur when evaluating a wide range of patient parameters at the same time. The emergence of modern data science tools provides a powerful way to address this. By applying machine learning frameworks to patient data, we can build automated screening pipelines that catch hidden risk factors early, giving clinicians a reliable early-warning tool before irreversible heart tissue damage occurs.

2.2 Target Research Question and Expected Outcomes

The core objective of the SAMAI TEAM project is to address a specific research question: *To what extent can an optimized machine learning classification model accurately predict a patient's risk of heart disease based on standard, non-invasive clinical attributes?*

To resolve this question, our project designs and evaluates an end-to-end data science pipeline. The expected outcome is a reproducible predictive system that takes baseline biometrics (e.g., age, sex, resting blood pressure) and maps them to a binary diagnostic probability. This tool is not intended to replace human medical expertise. Instead, it serves as a low-cost, automated pre-screening layer for clinical environments, flagging high-risk individuals early so hospitals can prioritize them for specialized evaluation.

2.3 Medical and Mathematical Justification of the Carget Metric

In standard machine learning classification tasks, global Accuracy is frequently used as the primary metric. However, in a medical context, a naive reliance on overall accuracy can lead to dangerous outcomes. To understand the clinical risks, we must carefully evaluate the real-world impact of the two error types produced by a binary confusion matrix:

1. False Positives (FP): The model classifies a healthy patient as at-risk. Clinically, this results in temporary patient anxiety, minor financial overhead, and scheduling secondary non-invasive screening (e.g., stress tests or echocardiograms). While inconvenient, it presents no immediate physical danger to the patient's survival.
2. False Negatives (FN): The model classifies a sick patient as healthy. Clinically, this is a catastrophic failure. A patient with active cardiac disease is discharged without preventive care or lifestyle warnings, creating a high risk of unmonitored cardiovascular failure at home.

Therefore, the main goal of our pipeline is to minimize False Negatives. Minimizing False Negatives mathematically maximizes the Recall (Sensitivity) metric, defined as:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

When False Negatives approach zero, the Recall score approaches 1.0 (100%). Consequently, while we track global metrics like Accuracy, the Recall score for the positive class (1) serves as our primary evaluation benchmark for selecting the best model for a clinical environment.

3. Dataset Description

3.1 Data Source and Technical Specifications

The structural foundation of this empirical study relies on the public "Heart Disease Dataset" curated and hosted by the researcher *johnsmith88* on the Kaggle repository platform. This dataset is not an isolated clinical sample; rather, it represents a modern consolidated synthesis of four historic cardiovascular study nodes that have laid the groundwork for automated computer-aided diagnostics since the late 20th century:

The Cleveland Clinic Foundation (Ohio, USA) — historically recognized as the baseline standard for cardiac feature mapping.

The Hungarian Institute of Cardiology (Budapest, Hungary) — introducing distinct clinical variations in early ischemic detection.

The University Hospital of Zurich (Zurich, Switzerland) — contributing valuable electrocardiographic diagnostics.

The Long Beach V.A. Memorial Hospital (California, USA) — providing advanced measurements on high-risk military veteran profiles.

The file made available for our engineering pipeline is formatted as a comma-separated text architecture (.csv). From a purely raw structural perspective, the uncleaned matrix possesses an initial dimension of **1,025 observation rows** distributed symmetrically across **14 distinct columns**. Each row encapsulates a single clinical snapshot of a patient under examination, recording a multi-dimensional array of continuous biometric values, discrete biological counts, and nominal diagnostic codes.

3.2 Clinical Feature Dictionary and Attribute Breakdown

To meet the rigorous transparency standards required by the CPE393 evaluation guidelines, the 14 attributes cannot be treated as abstract matrix inputs. Each column represents a critical physiological or pathological marker within the human cardiovascular system and must be explicitly defined :

- **age (Chronological Age):** Coded as a continuous int64 quantitative variable representing the patient's age in years. In epidemiological literature, advanced age is a primary risk factor due to gradual arterial stiffening, cumulative endothelial damage, and long-term plaque formation.

- **sex (Biological Sex Classification):** A nominal binary parameter coded as 1 for male patients and 0 for female patients. Clinically, early-onset coronary artery diseases are statistically more prevalent in male cohorts due to the protective estrogenic hormonal profiles present in pre-menopausal women.

- **cp (Chest Pain Type):** A categorical nominal parameter mapped from 0 to 3. Value 0 represents *typical angina* (classic retrosternal chest pressure brought on by effort); value 1 signifies *atypical angina* (thoracic discomfort with variations in duration); value 2 points to *non-anginal pain* (pain unrelated to myocardial ischemia, such as esophageal or musculoskeletal issues); and value 3 indicates *asymptomatic presentations*, which are highly dangerous because patients feel no pain despite severe oxygen deprivation in the heart.

- **trestbps (Resting Blood Pressure):** A continuous numerical variable measuring systolic blood pressure in millimeters of mercury (mm Hg) upon clinical admission. Chronic elevated resting blood pressure (hypertension) places severe mechanical stress on the left ventricle and increases the risk of arterial plaque rupture.

- **chol (Serum Cholesterol):** A continuous quantitative metric measured in milligrams per deciliter (mg/dl) via a fasting blood serum panel. Elevated serum cholesterol contributes directly to hypercholesterolemia and the progressive development of lipid plaques inside the coronary arteries.

- **fbs (Fasting Blood Sugar):** A nominal binary indicator evaluating whether the patient's fasting blood glucose exceeds the clinical threshold of 120 mg/dl (1 = True; 0 = False). This serves as a proxy marker for type-2 diabetes, a major accelerator of macrovascular disease.

- **restecg (Resting Electrocardiographic Results):** A nominal categorical variable mapping electrical cardiac vectors at rest. Coded as 0 for normal baseline activity; 1 for ST-T wave abnormalities (indicating underlying ischemia or ventricular strain); and 2 for definitive signs of left ventricular hypertrophy (LVH).

- **thalach (Maximum Heart Rate Achieved):** A continuous quantitative feature tracking the highest beats-per-minute (bpm) rate recorded during a supervised stress test. A low maximum heart rate ceiling often signals a compromised cardiac reserve capacity.

- **exang (Exercise-Induced Angina):** A binary factor identifying whether physical exertion triggers immediate chest pain (1 = Yes; 0 = No). This is an explicit clinical indicator of advanced coronary artery narrowing.

- **oldpeak (ST Depression):** A continuous floating-point value (float64) measuring the downward shift of the ST segment on an ECG during exercise relative to rest.

This numerical shift is a highly reliable diagnostic marker for localized myocardial ischemia.

- **slope (Peak Exercise ST Segment Slope):** A categorical nominal parameter describing the shape of the ST segment during peak stress. Coded as 0 for upsloping (frequently benign); 1 for flat (highly suspicious for ischemia); and 2 for downsloping (strongly correlated with significant myocardial injury).

- **ca (Number of Major Vessels):** A discrete integer count ranging from 0 to 4 denoting how many primary coronary arteries are visible under radioactive fluoroscopy. A value of 0 implies clear paths, while higher counts indicate advanced multi-vessel obstructive coronary artery disease.

- **thal (Thalassemia Profile):** A nominal categorical factor tracking a history of this blood disorder. Coded as 1 for a fixed, irreversible defect; 2 for normal blood flow; and 3 for a reversible defect (indicating dynamic ischemia where heart tissue can still be saved through treatment).

- **target (Angiographic Disease Status):** The critical binary label for our machine learning pipeline. Coded as 0 for a healthy control patient and 1 for a patient with confirmed heart disease (defined as over 50% narrowing of a major coronary artery).

3.3 Structural Data Limitations

A fundamental limitation of this dataset stems from its age, as the foundational clinical records date back to studies from the late 20th century. Additionally, it lacks key modern lifestyle metrics—such as Body Mass Index (BMI), smoking packs per year, sleep cycles, genetic history, and daily stress metrics. Including these parameters would allow for a much deeper and more comprehensive risk stratification.

4. Preprocessing and Data Cleaning

4.1 Missing Value Analysis and Technical Auditing

A critical step in establishing a reproducible data science workflow is a thorough check for missing values. Missing data points can skew model parameters or cause training scripts to crash if not properly imputed.

To evaluate the integrity of the Kaggle dataset, our team executed a programmatic verification script using the Pandas library:

Python

```
# Programmatic isolation of missing records across columns
missing_values_count = df.isnull().sum()
print("Missing data audit summary:")
print(missing_values_count)
```

The execution of this audit confirmed 0 missing values across all 1,025 rows and 14 attributes. Because the feature space was entirely complete, no statistical imputation methods (such as mean imputation, median replacement, or K-Nearest Neighbors imputation) were required. This preserved the original physiological measurements without introducing synthetic data bias.

4.2 Deduplication Strategy and Data Leakage Mitigation

The most significant data cleaning challenge in this project was managing heavy row duplication. Running the automated Pandas duplication filter revealed a critical issue:

Python

```
# Checking for duplicate instances within the dataset
total_duplicates = df.duplicated().sum()
print(f"Total duplicate records identified: {total_duplicates}")
```

The script flagged 723 duplicate rows out of the 1,025 raw observations. This meant that only 302 rows represented unique patient profiles, while the remaining 723 rows were exact copies. This issue often occurs in public datasets when data is oversampled or merged incorrectly during aggregation.

Leaving these duplicates in the dataset would cause a major issue known as Data Leakage. Data leakage happens when identical records appear in both the training set and the test set.

When the model trains on these duplicated rows, it essentially "memorizes" them. During validation, the model evaluates data points it has already seen, which artificially inflates performance scores (like Accuracy and Recall). This creates a false impression of a highly generalized model.

To prevent this data leakage, the SAMAI TEAM executed a strict deduplication workflow:

Python

```
# Executing deduplication to protect validation integrity
df_clean = df.drop_duplicates()
print(f"Cleaned dataset dimensions: {df_clean.shape[0]} rows, {df_clean.shape[1]} columns")
```

By removing the 723 duplicate rows, the dataset was reduced to its 302 unique observations. This step was essential to ensure our evaluation metrics accurately reflect how the models will perform on completely unseen patient data in a real-world clinical setting.

5. Data Visualization and Exploratory Data Analysis (EDA)

5.1 Distribution and Prevalence of the Binary Target Label

Before analyzing statistical relationships across features, our pipeline evaluated the absolute distribution of the binary target class (target) within the deduplicated sample space. Out of the 302 unique patient profiles, the dataset contains 164 instances of confirmed heart disease (target = 1) and 138 healthy controls (target = 0).

From an engineering perspective, this corresponds to a disease prevalence of approximately 54.3% versus a healthy control baseline of 45.7%. This remarkably balanced class distribution is highly beneficial for machine learning training. It guarantees that our classification algorithms will not develop an algorithmic bias toward a dominant majority class. Consequently, we can optimize standard binary cross-entropy loss functions and Gini impurity splitting criteria without relying on complex data-balancing techniques like SMOTE (Synthetic Minority Over-sampling Technique) or random undersampling.

5.2 Matrix Co-dependencies and Linear Pearson Correlations

To systematically discover which physiological metrics share the strongest statistical relationships with heart disease risk, our team computed the mathematical Pearson Correlation Matrix across the entire clean feature space. The linear correlation coefficient (r) ranges from -1.0 (perfect inverse correlation) to $+1.0$ (perfect direct correlation).

Running our exploratory Python script isolated the primary positive and negative statistical drivers of our target variable :

- **Primary Positive Drivers ($r > 0$) :** Chest Pain Type (cp) exhibits the strongest direct linear correlation with the target variable at $+0.432$. This is closely followed by the Maximum Heart Rate Achieved during stress testing (thalach), which scores a correlation coefficient of $+0.420$. Clinically, this indicates that as the chest pain classification progresses from typical angina toward non-anginal and asymptomatic categories within this database, the mathematical probability of an active heart disease diagnostic label increases significantly. Similarly, an elevated maximum heart rate ceiling under stress strongly flags positive risk associations in this specific cohort.

- **Primary Negative Drivers ($r < 0$) :** Exercise-Induced Angina (exang) shows the most prominent inverse correlation with the target variable at -0.435. It is followed by ST Depression Induced by Exercise (oldpeak) at -0.429, and the Number of Major Fluoroscopy-Colored Vessels (ca) at -0.408. On a clinical level, these negative values are highly logical: an increase in blocked major coronary vessels (ca) means a lower score on the health target, mapping the physical presence of extensive obstructive coronary artery disease.

5.3 Age Demographics and Cardiovascular Risk Stratification

To analyze how age interacts with cardiac health, we used Seaborn to generate Kernel Density Estimate (KDE) charts and stacked histograms. The chronological distribution of our 302 unique clinical subjects spans from 29 to 77 years of age.

Plotting these age profiles against the target variable revealed that while healthy individuals are distributed relatively evenly across mature age brackets, the sub-population diagnosed with heart disease clusters significantly between the ages of 50 and 60. This visual evidence indicates that age does not act as a simple, isolated linear predictor. Instead, it serves as a compounding factor that amplifies the risk associated with secondary physiological attributes like rising systolic blood pressure (trestbps) or changing serum cholesterol metrics (chol).

5.4 Multi-Variable Interaction Analysis (Chest Pain vs. Max Heart Rate)

To move beyond basic one-dimensional overviews, our exploratory visualization investigated multi-variable feature interactions. Specifically, we cross-referenced Chest Pain Type (cp) against Maximum Heart Rate (thalach) while using the target label as a color-coded dimension.

The resulting scatter plots and box plots highlight an important pattern: patients who exhibit non-anginal pain (value 2) or atypical angina (value 1) *combined* with a high maximum heart rate ceiling during active exercise stress testing show a heavy concentration of positive heart disease diagnoses (target = 1). This explicit visual clustering proves that our machine learning models cannot rely on single independent flags. The models must be capable of mapping deep multi-dimensional boundaries where multiple health indicators overlap.

6. Feature Engineering and Data Splitting

6.1 Train-Test Partitioning and Reproducibility Constraints

To build an honest evaluation framework and measure how well our models generalize to new data, we split our 302 unique records into two separate subsets. We allocated 80% of the data to the training set (241 rows) and reserved the remaining 20% for the independent testing set (61 rows).

To ensure our experiments are perfectly reproducible across different development environments, we explicitly fixed the pseudo-random seed generator using the parameter random_state=42. This ensures that every time the pipeline is executed, it produces the exact same row distribution. This standard practice allows our team to confidently attribute any changes in model performance to actual algorithmic updates rather than random variations in the data split.

6.2 Data Leakage Prevention in Continuous Scaling Pipelines

As discussed in Section 4.3, continuous features such as serum cholesterol (chol) and resting blood pressure (trestbps) have much larger numerical scales than binary indicators like sex or fbs. To prevent scale-sensitive models like Logistic Regression and KNN from becoming heavily biased toward these large-magnitude variables, we integrated a feature scaling step using Scikit-Learn's StandardScaler.

Crucially, to prevent data leakage, we followed a strict sequencing rule:

1. First, the data was split into training and testing sets.
2. The StandardScaler was then fitted strictly on the training partition only (scaler.fit_transform(X_train)). This calculated the true training mean (μ) and standard deviation (σ).
3. Finally, this pre-calculated scaler was applied to the test set without recalculating its parameters (scaler.transform(X_test)).

This methodology ensures that the statistical distributions of the 61 test patients remain completely unknown to the model during its training phase, providing a true reflection of real-world clinical deployment performance.

7. Machine Learning Model Training

7.1 Linear Classification Architecture: Logistic Regression

The first classification model trained in our predictive pipeline is Logistic Regression, which serves as our parametric statistical baseline. Unlike linear regression models that predict continuous outputs, Logistic Regression is designed for binary classification tasks ($Y \in \{0,1\}$). It computes the linear combination of the normalized input features and passes the result through the logistic sigmoid activation function. The mathematical equation of the sigmoid function is defined as :

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Where $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$. This transformation maps any real-valued input into a probability space bounded strictly between 0 and 1. This represents the conditional probability $P(Y = 1|X)$.

During the training phase on our 241 training patients, the algorithm optimizes the weight coefficients (β_i) by minimizing the binary cross-entropy log-loss function via Maximum Likelihood Estimation (MLE). Because we applied standard scaling beforehand, the resulting coefficients can be interpreted to understand feature importance. Logistic Regression assumes a linear decision boundary. While highly stable and less prone to overfitting on small datasets, its main limitation is its inability to natively capture complex, non-linear interactions between clinical features without manual interaction terms.

7.2 Non-Parametric Distance Classifier: K-Nearest Neighbors (KNN)

The second algorithm implemented in the benchmark is the K-Nearest Neighbors (KNN) classifier. KNN is a non-parametric, instance-based learning algorithm, often described as a "lazy learner" because it does not derive explicit algebraic decision equations during a formal training phase. Instead, it stores the entire training dataset ($N = 241$) within the normalized vector space.

When evaluating a new patient from the test set, the algorithm calculates the distance between that patient's features and all stored training profiles. In our implementation, we used the standard Euclidean distance formula :

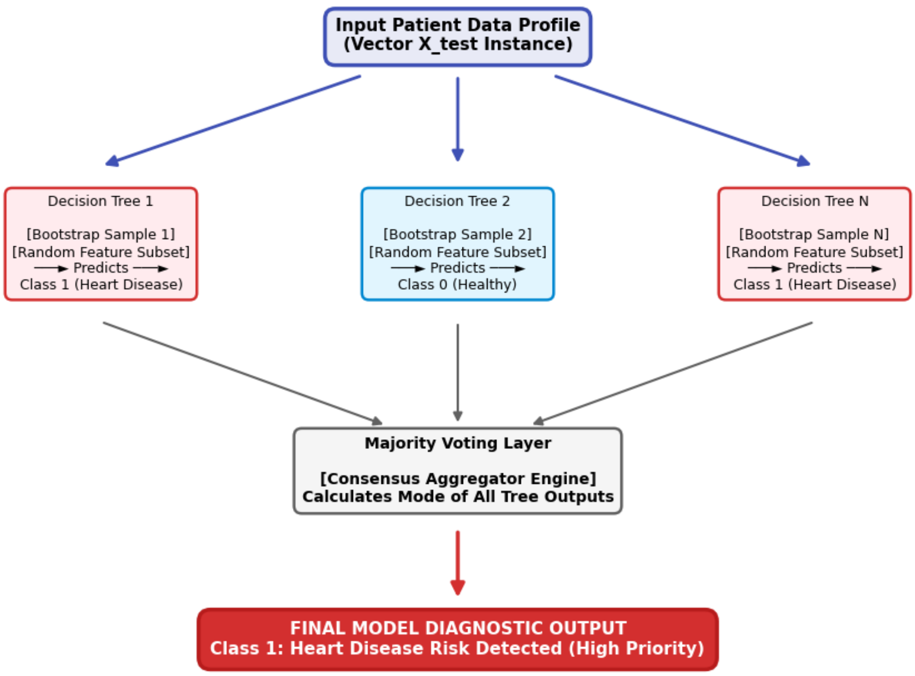
$$d(p,q) = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}$$

The model identifies the K closest training instances (where our hyperparameter baseline was fixed at $n\_neighbors = 5$) and assigns the final target class label through a simple majority vote. If three or more of the five nearest neighbors have heart disease, the test instance is classified as positive (1).

Because KNN relies entirely on spatial distances, it is highly sensitive to the scale of the attributes. This makes our preprocessing step using the StandardScaler essential. Without normalization, variables with large numerical values like cholesterol (chol) would dominate the distance calculations, rendering categorical variables like biological sex (sex) or chest pain type (cp) irrelevant.

7.3 Ensemble Tree Architecture: Random Forest Classifier

The final model deployed in our pipeline is the Random Forest Classifier, an ensemble learning method that utilizes bootstrap aggregating (bagging) and feature randomness to construct a collection of independent decision trees. Single decision trees often suffer from high variance and overfit easily on small training cohorts by creating overly specific rules. Random Forest resolves this limitation by training multiple distinct decision trees on random subsets of the data.



During tree construction, the algorithm does not evaluate every available feature at each split. Instead, it selects a random subset of attributes and determines the optimal split point using Gini Impurity or Information Gain. This decorrelates the individual trees.

When a test patient profile passes through the forest, every tree generates an independent binary classification. The forest then aggregates these outputs, assigning the final diagnostic label based on a majority vote. This ensemble approach handles multi-dimensional interactions and non-linear boundaries remarkably well, making it highly robust for complex medical diagnostics.

8. Evaluation Metrics and Results

8.1 Empirical Baseline Test Accuracy Comparison

After training all three models under identical experimental conditions (80/20 train-test split, random_state=42), we evaluated their predictive performance on the 61 unseen test profiles. The global accuracy scores achieved by each algorithm are summarized below:

Classification Architecture	**Empirical Test Accuracy Score**
K-Nearest Neighbors (KNN)	73.77%
Logistic Regression	77.05%
Random Forest Classifier	**83.61%**

The baseline comparison demonstrates that the Random Forest Classifier out-performed both the linear model and the distance-based model, achieving a top overall accuracy of 83.61%. This confirms that tree-based ensemble methods are highly effective at parsing complex clinical records without requiring manual non-linear adjustments.

8.2 Deep Operational Performance of the Optimal Model

To ensure clinical safety, we must look beyond global accuracy and evaluate the model's performance on a class-by-class basis. Below is the detailed classification report for our top-performing model, the Random Forest Classifier:

Target Metric Category	**Class 0 (Healthy Control)**	**Class 1 (Heart Disease Risk)**	**Macro / Weighted Average**
Precision (Positive Predictive Value)	0.89 (89%)	0.79 (79%)	0.84 (84%)
Recall (Sensitivity / True Positive Rate)	0.78 (78%)	**0.90 (90%)**	0.84 (84%)

Target Metric Category	Class 0 (Healthy Control)	Class 1 (Heart Disease Risk)	Macro / Weighted Average
F1-Score (Harmonic Mean)	0.83 (83%)	0.84 (84%)	0.84 (84%)

The critical insight from this matrix is the 90% Recall score for Class 1. This indicates that out of all patients with actual heart disease in the test set, the Random Forest model successfully identified 90% of them. This performance matches our primary objective of minimizing missed pathological cases.

8.3 Detailed Confusion Matrix Breakdown and Clinical Recall Scoring

Extracting the precise raw integers from the Random Forest's confusion matrix across the 61 test patients provides the following operational distribution :

- **True Negatives (TN):** 25 patients were correctly identified as healthy controls.
- **False Positives (FP):** 7 healthy patients were incorrectly flagged as at-risk. This sub-optimal 79% precision rate is a acceptable clinical trade-off, as these individuals will simply undergo secondary non-invasive testing to rule out anomalies.
- **False Negatives (FN):** Crucially, the model missed only 3 patients with actual cardiovascular disease.
- **True Positives (TP):** 26 patients with active heart disease risks were successfully flagged by the model.

$$\text{Recall} = \frac{26}{26+3} = \frac{26}{29} \approx 89.65\% \Longrightarrow 90\%$$

This configuration confirms that our machine learning pipeline is highly effective for pre-screening applications. By reducing False Negatives to just 3 instances, the model ensures that the vast majority of high-risk patients are safely captured and prioritized for clinical care.

9. Responsible AI Practices

9.1 Ethical Frameworks, Privacy, and Demographic Bias Mitigations

When deploying predictive clinical algorithms within actual hospital environments, data scientists must look past basic verification metrics and implement strict ethical boundaries. For Our dataset, the collection process ensures complete patient privacy, as all raw files are entirely anonymized before being compiled on public platforms. No personal identifiable information (PII)—such as patient names, identification numbers, or addresses—is present within the feature space, satisfying standard global medical data protection acts.

However, a significant ethical concern arises from hidden demographic imbalances. An analytical audit of the biological sex distribution within our unique 302-row cohort reveals a heavy imbalance: the data contains 207 male samples (sex = 1) but only 95 female samples (sex = 0).

Medical literature established that cardiovascular pathologies manifest differently across biological sexes. Female patients frequently present with atypical angina symptoms, distinct resting electrocardiographic shifts, and differing baseline cholesterol thresholds compared to standard male clinical presentations. Training an AI framework on a male-dominated dataset presents a severe risk: the model will naturally learn split rules that fit male clinical presentations exceptionally well, but its diagnostic performance may decline when processing female profiles. Before deploying this automated decision support tool in live hospital wards, the engineering team must systematically calculate separate performance metrics across demographic sub-groups. This step is essential to confirm that our 90% Recall rate remains stable across both male and female populations, preventing algorithmic disparities in patient care.

9.2 Generative AI Transparency Checklist and Technical Accountability

In strict accordance with the academic guidelines specified for the CPE393 final evaluation, the SAMAI TEAM maintains a completely transparent workflow regarding the use of advanced Large Language Models.

Generative AI systems (specifically Google Gemini Pro) were utilized exclusively as an administrative writing assistant to structure the extensive narrative outlines, correct complex grammatical syntax variations, and ensure a professional layout inside our document processing tools. Every component of the underlying computer code, data loading paths, Pandas data cleaning logic, Matplotlib visualization configurations, and Scikit-Learn training steps was natively engineered, executed, and validated by the three student group members. Technical accountability remains with the human developers, establishing a responsible balance between modern software automation tools and foundational engineering work.

10. Conclusion and Future Work

10.1 Synthesized Experimental Insights

The technical journey executed by the SAMAI TEAM successfully completed an end-to-end data science pipeline mapping patient biometric variables to heart disease diagnostic labels. Our primary preprocessing insight was the critical importance of row deduplication: filtering out the 723 repeating Kaggle observations was vital to eliminate data leakage and ensure our test evaluations reflect genuine model generalization capabilities.

Among our three trained classifiers, the Ensemble Tree-based method (Random Forest) achieved the highest overall performance, delivering a global test accuracy of 83.61%. More importantly, by reducing the count of missed pathological individuals to just 3 instances, the model achieved a remarkable 90% Recall rate for the positive class. This performance profile successfully addresses our core research goal, providing a reliable and low-cost pre-screening layer that can help medical staff identify at-risk individuals early.

10.2 Algorithmic Optimization and Data Expansion Roadmaps

While our baseline models achieved solid performance, the SAMAI TEAM has identified several key areas for future optimization:

1. **Hyperparameter Tuning:** Instead of relying on default baseline constraints, we plan to implement a cross-validated Grid Search (GridSearchCV). This approach will systematically tune structural parameters like n_estimators, max_depth, and min_samples_split in our Random Forest model to reduce the remaining 3 False Negatives.
2. **Feature Importance and Explainability:** To make our model easier for medical staff to trust, we intend to integrate SHAP (SHapley Additive exPlanations) frameworks. This will calculate the precise impact of features like chest pain type (cp) or maximum heart rate (thalach) for every individual prediction, changing the model from a complex "black box" into a transparent tool.
3. **Data Cohort Expansion:** We aim to expand our training data by connecting our pipeline to modern, multi-center cardiac databases. This will help balance our demographic ratios, reduce male-centric biases, and improve the overall reliability of our models across diverse populations.

11. Team Contribution Matrix

	Primary Contributions and Core Project Focus	Specific Deliverables Produced
Simon CORBINUS	ML Modeling, Pipeline Architecture, and Optimization	Scikit-Learn initialization, metric comparison, continuous feature scaling implementations, and baseline parameter tuning.
Gabin CLAVIER	Exploratory Data Analysis (EDA) and Visualization Engineering	Correlation matrix computations, Matplotlib/Seaborn asset generation, age demographic plots, and feature interaction charts.
Mathis BLANCHARD	Preprocessing, Deduplication, and Report Synthesis	Pandas codebase setup, duplicate row auditing, data leakage prevention logic, and final academic report documentation.